MODULE 06 · DOMAIN 1 - MONITORING, LOGGING, ANALYSIS, REMEDIATION, AND PERFORMANCE OPTIMIZATION

EventBridge and Event Routing

Buses, patterns, targets - and the six reasons a rule quietly does not fire.

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EventBridge routes on the **shape** of an event. A rule that does not fire is nearly always a pattern that does not match, and nothing tells you.

The event envelope

- Every event has the same outer fields: **version, id, detail-type, source, account, time, region, resources, detail**.
- **source** identifies the producer: **aws.ec2**, **aws.s3**, or your own reverse-DNS style name.
- **detail** is the service-specific payload. Everything above it is standard, which is why one pattern language works for all events.
- It is **detail-type** with a hyphen, and matching is **case-sensitive**. Both catch people constantly.

> ON THE EXAM
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> Maximum event size is 256 KB. Larger payloads go to S3 and the event carries a reference. That is the claim-check pattern.

Buses, and which one you are on

- **The default bus** receives every AWS service event in the account automatically. You do not subscribe to anything.
- **Custom buses** are for your own application events, giving separate rules, permissions and quotas.
- **Partner buses** receive events from SaaS providers such as payment or monitoring vendors.
- **Scheduled rules only work on the default bus.** Creating one on a custom bus does not run.

> ON THE EXAM
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> AWS service events land on the default bus and nowhere else. A rule for them on a custom bus will never match anything.

Pattern matching, beyond exact values

OPERATOR	WHAT IT DOES
`Exact value`	"source": ["aws.ec2"] - an array of values means OR, not AND
`prefix / suffix`	Match the start or end of a string, such as a bucket name prefix
`anything-but`	Everything except the listed values. Useful for excluding a known-good case
`numeric`	Comparisons such as greater than or between, on numbers in the detail
`exists`	Whether a field is present at all, regardless of value
`cidr`	Match an IP address against a network range
`equals-ignore-case, wildcard`	Case-insensitive matching, and * within a string

ON THE EXAM

An array is OR. To require two conditions, put them in two different fields. There is no AND inside one field.

Four pattern rules that decide questions

- **The pattern must mirror the event's structure.** A nested field has to be nested identically in the pattern.
- **A field you omit is not filtered.** Leave out source and you match every source, which is usually not what you meant.
- **Matching is case-sensitive** unless you use equals-ignore-case.
- For an **array in the event**, the pattern matches if **any element** matches.

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> Copy a real event from CloudTrail or the sandbox and build the pattern from it. Writing one from memory is how afternoons disappear.

Why a rule does not fire: configuration

- **Wrong bus.** AWS service events are on the default bus. Scheduled rules must be on the default bus.
- **Pattern does not match.** Test it against a real event with the sandbox before blaming anything else.
- **Wrong Region.** Rules are regional. Global service events such as IAM appear in us-east-1.
- **The event was never produced.** S3 object-level events need EventBridge notifications enabled on the bucket, or CloudTrail data events.

> ON THE EXAM
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> Nothing is logged when a pattern fails to match. Silence is the normal outcome, not a symptom of a fault.

Why a rule does not fire: delivery

- **The rule is disabled.** Obvious, and worth ten seconds of checking before an hour of theorising.
- **The target rejects the invocation.** EventBridge needs permission: a resource policy on SQS, SNS or Lambda, or an IAM role for others.
- Here the rule **did** match, so **FailedInvocations** increments. That metric is how you tell the two halves apart.
- An **encrypted SQS queue** also needs the KMS key policy to allow EventBridge to Decrypt and GenerateDataKey.

> ON THE EXAM
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> Matched but undelivered shows in FailedInvocations. Never matched shows nowhere. Check that metric first and you halve the search space.

Targets, and reshaping what they receive

- **Up to 5 targets per rule**, and that limit is firm. More consumers means more rules, or fan out through SNS.
- Targets include Lambda, Step Functions, SQS, SNS, ECS tasks, **SSM Automation and Run Command**, Kinesis, another event bus.
- The **input transformer** reshapes the event before it arrives, so the target gets the fields it expects.
- **API destinations** deliver to any HTTPS endpoint, with managed authentication and rate limiting.

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> SSM Automation as a direct target means many remediations need no Lambda function at all. That matters for "least operational overhead".

When delivery fails

- Default retry policy: **24 hours and up to 185 attempts**, with exponential backoff and jitter.
- After that the event is **dropped silently** unless a dead-letter queue is configured.
- The DLQ is a **standard SQS queue**, set **per target**, and the message carries the error code and rule ARN.
- EventBridge retries **5xx and 429** responses. It does **not** retry other 4xx errors, because they will not improve.

> ON THE EXAM
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> Watch FailedInvocations and InvocationsSentToDlq. Without a DLQ, a broken target loses events with nothing to show for it.

Archive and replay

- An **archive** stores events arriving on a bus, optionally filtered by pattern, for a retention period you choose.
- **Replay** re-sends archived events over a chosen time range to the bus, and rules process them again.
- Recovers from a bug in a downstream consumer, or a target that was broken for hours.
- Replay is **not** a dead-letter queue. It replays what the bus received, not what failed delivery.

> ON THE EXAM
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> "Reprocess yesterday's events after fixing the consumer" is replay. "Recover the specific events that failed" is the DLQ.

Scheduled rule or EventBridge Scheduler?

SCHEDULED RULE

The original, now legacy

- cron or rate, on the default bus only
- UTC only, no time zone awareness
- Targets limited to EventBridge target types
- Fine for something you already have

EVENTBRIDGE SCHEDULER

What AWS now recommends

- Time zones and daylight saving handled
- One-time schedules as well as recurring
- Over 270 services as direct targets
- Flexible time windows to spread load

THE TIEBREAKER

AWS explicitly recommends Scheduler for new work. Time zones, one-time schedules or huge scale each decide it on their own.

EventBridge Pipes: point to point

- Connects **one source to one target**, with optional **filtering** and **enrichment** in between.
- Sources are polling sources: **DynamoDB Streams, Kinesis, SQS, Amazon MQ, MSK** and self-managed Kafka.
- Enrichment can call **Lambda, Step Functions, API Gateway or an API destination** to add context before delivery.
- A bus is **one to many** with rules. A pipe is **one to one**. That is the distinction the exam wants.

> ON THE EXAM
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> Pipes exists to delete the glue Lambda that only polled a stream, reshaped a record and called something else.

EventBridge or SNS?

EVENTBRIDGE WHEN

Routing on content

- Rich filtering on the whole event body
- Hundreds of AWS services as native sources
- Archive, replay and schema registry
- Slightly higher latency, around half a second

SNS WHEN

Fan-out at speed

- Very high throughput, very low latency
- Filtering on message attributes, not full content
- Many subscribers of the same message
- SMS, email and mobile push endpoints

THE TIEBREAKER

They combine well. EventBridge routes and filters; SNS fans a single routed event out to many subscribers.

Across accounts and Regions

- Target **another account's event bus ARN** directly. Cross-Region works the same way, using the full ARN.
- The **receiving bus needs a resource policy** allowing events:PutEvents from the sending account or organisation.
- Centralise by having every account forward to a bus in a security or operations account.
- **Global endpoints** use a Route 53 health check to fail events over to a secondary Region automatically.

> ON THE EXAM
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> Sending is a rule target; receiving is a resource policy. Forget the second and events vanish without an obvious error.

A CloudOps engineer creates a custom event bus named ops-events and adds a rule on it with the pattern {"source": ["aws.ec2"], "detail-type": ["EC2 Instance State-change Notification"]}, targeting a Lambda function that logs terminations.

Instances are terminated repeatedly over two days. The Lambda function is never invoked. CloudWatch shows no Invocations for the function and no FailedInvocations for the rule.

What is the cause?

A The Lambda function's resource policy does not allow EventBridge to invoke it

B AWS service events are delivered to the default event bus, not to custom buses

C The detail-type string is case-sensitive and does not match

D CloudTrail data events must be enabled for EC2 state changes to be published

◆ PAUSE AND DECIDE

What is the cause?

A Lambda resource policy

A permissions failure means the rule matched and delivery failed, which would increment FailedInvocations. The scenario says it is zero.

B Service events go to the default bus

The rule is on a custom bus, so no EC2 event ever reaches it. Nothing matches, nothing fails, and both metrics stay at zero.

C Case-sensitive mismatch

The string quoted is the correct one. Plausible in general, but this particular pattern is right.

D CloudTrail data events

EC2 state changes are native EventBridge events. They do not depend on CloudTrail at all.

THE SIGNAL

FailedInvocations at zero is the discriminator. It proves the rule never matched, which rules out every permissions answer.

> A rule on the default bus matches a nightly batch completion event and targets an SQS queue encrypted with a customer managed KMS key. The rule's FailedInvocations metric increments every night, and no messages arrive in the queue.
>
> The queue's resource policy already allows events.amazonaws.com to call sqs:SendMessage.

What is the most likely remaining cause?

A The rule needs a dead-letter queue configured before it can deliver

B The KMS key policy does not allow EventBridge to call Decrypt and GenerateDataKey

C The event pattern does not match, so nothing is delivered

D The rule has exceeded the limit of 5 targets

◆ PAUSE AND DECIDE

What is the most likely remaining cause?

A **Needs a DLQ first**

A DLQ catches events after retries are exhausted. It never enables delivery, and its absence causes no failure.

B **KMS key policy missing permissions**

Writing to an encrypted queue needs GenerateDataKey and Decrypt on the key, in addition to SendMessage on the queue.

C **Pattern does not match**

Ruled out by the evidence. FailedInvocations only increments when a rule matched and delivery then failed.

D **Too many targets**

Exceeding five targets is rejected when you configure the rule, not silently at delivery time each night.

THE SIGNAL

The mirror image of scenario 1. FailedInvocations incrementing proves the pattern is fine and the problem is downstream.

A company runs a compliance report every weekday at 07:00 in London local time, currently triggered by a scheduled EventBridge rule with a cron expression. Twice a year, when British Summer Time starts or ends, the report runs an hour early or an hour late for several days until someone notices.

The team also wants to add roughly 4,000 further schedules for per-customer report generation, spread across many time zones.

What should the team use?

A Keep scheduled rules and add a Lambda function that adjusts the cron expressions twice a year

B EventBridge Scheduler, with time zone aware schedules and flexible time windows

C An EventBridge Pipe with a Kinesis source triggered by a clock stream

D Scheduled rules on a custom event bus, one per customer time zone

◆ PAUSE AND DECIDE

What should the team use?

A. Lambda that rewrites cron twice a year

Rebuilds daylight saving by hand, in code nobody touches for six months. Maximum operational overhead, and it will fail quietly.

B. EventBridge Scheduler

Time zones and DST are native, it scales to millions of schedules, and flexible time windows stop 4,000 jobs firing on the same second.

C. Pipe with a Kinesis source

Pipes connects a source to a target. It is not a scheduler, and there is no clock stream to poll.

D. Scheduled rules on a custom bus

Doubly wrong: scheduled rules do not work on a custom bus at all, and this still does not solve daylight saving.

THE SIGNAL

Time zones, one-time schedules, or very large numbers of schedules. Any one of those three points at Scheduler on its own.

> An operations team must automatically restart a specific Windows service on any EC2 instance where a CloudWatch alarm reports the service has stopped. The instances are Systems Manager managed and an SSM Automation runbook already exists that performs the restart correctly.
>
> The team wants the least operational overhead and no code to maintain.

How should the team wire this up?

A. An EventBridge rule matching the alarm state change, targeting the SSM Automation runbook directly

B. An EventBridge rule targeting a Lambda function that calls StartAutomationExecution

C. An SNS topic subscribed to the alarm, with a Lambda subscriber that starts the runbook

D. An EventBridge Pipe with the alarm as the source and the runbook as the target

◆ PAUSE AND DECIDE

How should the team wire this up?

A Rule targets the runbook directly

SSM Automation is a native EventBridge target. No function, no code, no deployment pipeline for glue logic.

B Rule to Lambda to runbook

Works perfectly and is what most teams build first. The Lambda function does nothing EventBridge cannot do, and it still needs owning.

C SNS then Lambda

Adds a topic and a function. Two components to maintain where the requirement asked for the fewest.

D EventBridge Pipe

Pipes sources are polling sources such as queues and streams. An alarm state change is not one of them.

THE SIGNAL

"No code to maintain" is the phrase. When a native target exists, a Lambda function in the middle is a wrong answer that works.

The five things to carry forward

- **AWS service events and scheduled rules live on the default bus.** A custom bus is for your own events.
- **FailedInvocations tells you which failure you have.** Zero means it never matched; climbing means delivery failed.
- **An array in a pattern means OR.** An omitted field is not filtered at all.
- **24 hours, 185 retries, then dropped** unless a DLQ exists. Archive and replay is a different tool.
- **SSM Automation is a native target.** A Lambda function that only forwards is avoidable overhead.

> ON THE EXAM
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> Next: Systems Manager itself – the agent, Session Manager, Run Command, Patch Manager and State Manager.